GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 1

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 1	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Structure of the Atom

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.

Key Inquiry Question: How does the arrangement of subatomic particles help us describe an atom?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about structure of the atom. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the arrangement of subatomic particles help us describe an atom?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the arrangement of subatomic particles help us describe an atom? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 2

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 1	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Structure of the Atom

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.

Key Inquiry Question: How does the arrangement of subatomic particles help us describe an atom?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about structure of the atom. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the arrangement of subatomic particles help us describe an atom?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the arrangement of subatomic particles help us describe an atom? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 3

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 1	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Structure of the Atom

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.

Key Inquiry Question: How does the arrangement of subatomic particles help us describe an atom?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about structure of the atom. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the arrangement of subatomic particles help us describe an atom?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the arrangement of subatomic particles help us describe an atom? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 4

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 1	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Structure of the Atom

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.

Key Inquiry Question: How does the arrangement of subatomic particles help us describe an atom?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about structure of the atom. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the arrangement of subatomic particles help us describe an atom?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the arrangement of subatomic particles help us describe an atom? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 5

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 1	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Structure of the Atom

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.

Key Inquiry Question: How does the arrangement of subatomic particles help us describe an atom?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about structure of the atom. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the arrangement of subatomic particles help us describe an atom?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the arrangement of subatomic particles help us describe an atom? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 6

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 2	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Structure of the Atom

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.

Key Inquiry Question: How does the arrangement of subatomic particles help us describe an atom?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about structure of the atom. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the arrangement of subatomic particles help us describe an atom?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the arrangement of subatomic particles help us describe an atom? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 7

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 2	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Structure of the Atom

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.

Key Inquiry Question: How does the arrangement of subatomic particles help us describe an atom?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about structure of the atom. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the arrangement of subatomic particles help us describe an atom?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the arrangement of subatomic particles help us describe an atom? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 8

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 2	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Structure of the Atom

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.

Key Inquiry Question: How does the arrangement of subatomic particles help us describe an atom?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about structure of the atom. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the arrangement of subatomic particles help us describe an atom?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the arrangement of subatomic particles help us describe an atom? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 9

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 2	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Structure of the Atom

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.

Key Inquiry Question: How does the arrangement of subatomic particles help us describe an atom?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about structure of the atom. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the arrangement of subatomic particles help us describe an atom?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the arrangement of subatomic particles help us describe an atom? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 10

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 2	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Structure of the Atom

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.

Key Inquiry Question: How does the arrangement of subatomic particles help us describe an atom?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about structure of the atom. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the arrangement of subatomic particles help us describe an atom?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the arrangement of subatomic particles help us describe an atom? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 11

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 3	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Structure of the Atom

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.

Key Inquiry Question: How does the arrangement of subatomic particles help us describe an atom?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about structure of the atom. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the arrangement of subatomic particles help us describe an atom?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the arrangement of subatomic particles help us describe an atom? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 12

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 3	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Structure of the Atom

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.

Key Inquiry Question: How does the arrangement of subatomic particles help us describe an atom?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about structure of the atom. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the arrangement of subatomic particles help us describe an atom?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the arrangement of subatomic particles help us describe an atom? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 13

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 3	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Structure of the Atom

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.

Key Inquiry Question: How does the arrangement of subatomic particles help us describe an atom?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about structure of the atom. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the arrangement of subatomic particles help us describe an atom?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the arrangement of subatomic particles help us describe an atom? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 14

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 3	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Structure of the Atom

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.

Key Inquiry Question: How does the arrangement of subatomic particles help us describe an atom?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about structure of the atom. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the arrangement of subatomic particles help us describe an atom?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the arrangement of subatomic particles help us describe an atom? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 15

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 3	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Metals and Alloys

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate selected physical properties and uses of metals and alloys.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate safe handling, responsible use and care for materials and the environment.

Key Inquiry Question: Why are different metals and alloys selected for particular purposes?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about metals and alloys. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: Why are different metals and alloys selected for particular purposes?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: Why are different metals and alloys selected for particular purposes? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 16

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 4	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Metals and Alloys

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate selected physical properties and uses of metals and alloys.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate safe handling, responsible use and care for materials and the environment.

Key Inquiry Question: Why are different metals and alloys selected for particular purposes?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about metals and alloys. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: Why are different metals and alloys selected for particular purposes?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: Why are different metals and alloys selected for particular purposes? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 17

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 4	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Metals and Alloys

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate selected physical properties and uses of metals and alloys.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate safe handling, responsible use and care for materials and the environment.

Key Inquiry Question: Why are different metals and alloys selected for particular purposes?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about metals and alloys. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: Why are different metals and alloys selected for particular purposes?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: Why are different metals and alloys selected for particular purposes? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 18

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 4	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Metals and Alloys

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate selected physical properties and uses of metals and alloys.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate safe handling, responsible use and care for materials and the environment.

Key Inquiry Question: Why are different metals and alloys selected for particular purposes?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about metals and alloys. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: Why are different metals and alloys selected for particular purposes?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: Why are different metals and alloys selected for particular purposes? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 19

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 4	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Metals and Alloys

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate selected physical properties and uses of metals and alloys.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate safe handling, responsible use and care for materials and the environment.

Key Inquiry Question: Why are different metals and alloys selected for particular purposes?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about metals and alloys. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: Why are different metals and alloys selected for particular purposes?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: Why are different metals and alloys selected for particular purposes? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 20

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 4	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Metals and Alloys

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate selected physical properties and uses of metals and alloys.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate safe handling, responsible use and care for materials and the environment.

Key Inquiry Question: Why are different metals and alloys selected for particular purposes?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about metals and alloys. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: Why are different metals and alloys selected for particular purposes?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: Why are different metals and alloys selected for particular purposes? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 21

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 5	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Metals and Alloys

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate selected physical properties and uses of metals and alloys.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate safe handling, responsible use and care for materials and the environment.

Key Inquiry Question: Why are different metals and alloys selected for particular purposes?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about metals and alloys. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: Why are different metals and alloys selected for particular purposes?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: Why are different metals and alloys selected for particular purposes? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 22

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 5	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Metals and Alloys

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate selected physical properties and uses of metals and alloys.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate safe handling, responsible use and care for materials and the environment.

Key Inquiry Question: Why are different metals and alloys selected for particular purposes?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about metals and alloys. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: Why are different metals and alloys selected for particular purposes?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: Why are different metals and alloys selected for particular purposes? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 23

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 5	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Metals and Alloys

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate selected physical properties and uses of metals and alloys.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate safe handling, responsible use and care for materials and the environment.

Key Inquiry Question: Why are different metals and alloys selected for particular purposes?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about metals and alloys. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: Why are different metals and alloys selected for particular purposes?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: Why are different metals and alloys selected for particular purposes? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 24

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 5	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Metals and Alloys

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate selected physical properties and uses of metals and alloys.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate safe handling, responsible use and care for materials and the environment.

Key Inquiry Question: Why are different metals and alloys selected for particular purposes?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about metals and alloys. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: Why are different metals and alloys selected for particular purposes?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: Why are different metals and alloys selected for particular purposes? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 25

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 5	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Metals and Alloys

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate selected physical properties and uses of metals and alloys.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate safe handling, responsible use and care for materials and the environment.

Key Inquiry Question: Why are different metals and alloys selected for particular purposes?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about metals and alloys. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: Why are different metals and alloys selected for particular purposes?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: Why are different metals and alloys selected for particular purposes? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 26

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 6	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Metals and Alloys

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate selected physical properties and uses of metals and alloys.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate safe handling, responsible use and care for materials and the environment.

Key Inquiry Question: Why are different metals and alloys selected for particular purposes?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about metals and alloys. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: Why are different metals and alloys selected for particular purposes?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: Why are different metals and alloys selected for particular purposes? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 27

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 6	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Metals and Alloys

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate selected physical properties and uses of metals and alloys.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate safe handling, responsible use and care for materials and the environment.

Key Inquiry Question: Why are different metals and alloys selected for particular purposes?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about metals and alloys. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: Why are different metals and alloys selected for particular purposes?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: Why are different metals and alloys selected for particular purposes? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 28

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 6	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Metals and Alloys

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate selected physical properties and uses of metals and alloys.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate safe handling, responsible use and care for materials and the environment.

Key Inquiry Question: Why are different metals and alloys selected for particular purposes?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about metals and alloys. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: Why are different metals and alloys selected for particular purposes?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: Why are different metals and alloys selected for particular purposes? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 29

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 6	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Metals and Alloys

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate selected physical properties and uses of metals and alloys.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate safe handling, responsible use and care for materials and the environment.

Key Inquiry Question: Why are different metals and alloys selected for particular purposes?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about metals and alloys. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: Why are different metals and alloys selected for particular purposes?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: Why are different metals and alloys selected for particular purposes? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 30

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 6	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Metals and Alloys

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate selected physical properties and uses of metals and alloys.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate safe handling, responsible use and care for materials and the environment.

Key Inquiry Question: Why are different metals and alloys selected for particular purposes?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about metals and alloys. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: Why are different metals and alloys selected for particular purposes?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: Why are different metals and alloys selected for particular purposes? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 31

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 7	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Water Hardness

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate safe water practices and use resources responsibly.

Key Inquiry Question: How can we identify hard water and make it more suitable for use?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about water hardness. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How can we identify hard water and make it more suitable for use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How can we identify hard water and make it more suitable for use? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 32

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 7	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Water Hardness

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate safe water practices and use resources responsibly.

Key Inquiry Question: How can we identify hard water and make it more suitable for use?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about water hardness. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How can we identify hard water and make it more suitable for use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How can we identify hard water and make it more suitable for use? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 33

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 7	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Water Hardness

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate safe water practices and use resources responsibly.

Key Inquiry Question: How can we identify hard water and make it more suitable for use?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about water hardness. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How can we identify hard water and make it more suitable for use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How can we identify hard water and make it more suitable for use? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 34

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 7	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Water Hardness

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate safe water practices and use resources responsibly.

Key Inquiry Question: How can we identify hard water and make it more suitable for use?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about water hardness. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How can we identify hard water and make it more suitable for use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How can we identify hard water and make it more suitable for use? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 35

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 7	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Water Hardness

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate safe water practices and use resources responsibly.

Key Inquiry Question: How can we identify hard water and make it more suitable for use?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about water hardness. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How can we identify hard water and make it more suitable for use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How can we identify hard water and make it more suitable for use? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 36

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 8	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Water Hardness

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate safe water practices and use resources responsibly.

Key Inquiry Question: How can we identify hard water and make it more suitable for use?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about water hardness. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How can we identify hard water and make it more suitable for use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How can we identify hard water and make it more suitable for use? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 37

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 8	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Water Hardness

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate safe water practices and use resources responsibly.

Key Inquiry Question: How can we identify hard water and make it more suitable for use?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about water hardness. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How can we identify hard water and make it more suitable for use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How can we identify hard water and make it more suitable for use? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 38

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 8	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Water Hardness

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate safe water practices and use resources responsibly.

Key Inquiry Question: How can we identify hard water and make it more suitable for use?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about water hardness. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How can we identify hard water and make it more suitable for use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How can we identify hard water and make it more suitable for use? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 39

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 8	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Water Hardness

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate safe water practices and use resources responsibly.

Key Inquiry Question: How can we identify hard water and make it more suitable for use?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about water hardness. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How can we identify hard water and make it more suitable for use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How can we identify hard water and make it more suitable for use? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 40

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 8	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Water Hardness

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate safe water practices and use resources responsibly.

Key Inquiry Question: How can we identify hard water and make it more suitable for use?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about water hardness. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How can we identify hard water and make it more suitable for use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How can we identify hard water and make it more suitable for use? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 41

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 9	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Water Hardness

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate safe water practices and use resources responsibly.

Key Inquiry Question: How can we identify hard water and make it more suitable for use?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about water hardness. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How can we identify hard water and make it more suitable for use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How can we identify hard water and make it more suitable for use? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 42

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 9	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Water Hardness

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate safe water practices and use resources responsibly.

Key Inquiry Question: How can we identify hard water and make it more suitable for use?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about water hardness. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How can we identify hard water and make it more suitable for use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How can we identify hard water and make it more suitable for use? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 43

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 9	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Water Hardness

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate safe water practices and use resources responsibly.

Key Inquiry Question: How can we identify hard water and make it more suitable for use?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about water hardness. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How can we identify hard water and make it more suitable for use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How can we identify hard water and make it more suitable for use? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 44

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 9	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Mixtures, Elements and Compounds

Sub-strand: Water Hardness

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate safe water practices and use resources responsibly.

Key Inquiry Question: How can we identify hard water and make it more suitable for use?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about water hardness. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How can we identify hard water and make it more suitable for use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How can we identify hard water and make it more suitable for use? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 45

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 9	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain processes and requirements involved in plant nutrition.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate the importance of plants and care for the environment.

Key Inquiry Question: What conditions do green plants require to manufacture food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: What conditions do green plants require to manufacture food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: What conditions do green plants require to manufacture food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 46

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 10	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain processes and requirements involved in plant nutrition.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate the importance of plants and care for the environment.

Key Inquiry Question: What conditions do green plants require to manufacture food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: What conditions do green plants require to manufacture food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: What conditions do green plants require to manufacture food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 47

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 10	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain processes and requirements involved in plant nutrition.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate the importance of plants and care for the environment.

Key Inquiry Question: What conditions do green plants require to manufacture food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: What conditions do green plants require to manufacture food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: What conditions do green plants require to manufacture food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 48

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 10	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain processes and requirements involved in plant nutrition.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate the importance of plants and care for the environment.

Key Inquiry Question: What conditions do green plants require to manufacture food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: What conditions do green plants require to manufacture food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: What conditions do green plants require to manufacture food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 49

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 10	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain processes and requirements involved in plant nutrition.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate the importance of plants and care for the environment.

Key Inquiry Question: What conditions do green plants require to manufacture food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: What conditions do green plants require to manufacture food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: What conditions do green plants require to manufacture food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 50

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 10	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain processes and requirements involved in plant nutrition.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate the importance of plants and care for the environment.

Key Inquiry Question: What conditions do green plants require to manufacture food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: What conditions do green plants require to manufacture food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: What conditions do green plants require to manufacture food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 51

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 11	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain processes and requirements involved in plant nutrition.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate the importance of plants and care for the environment.

Key Inquiry Question: What conditions do green plants require to manufacture food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: What conditions do green plants require to manufacture food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: What conditions do green plants require to manufacture food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 52

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 11	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain processes and requirements involved in plant nutrition.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate the importance of plants and care for the environment.

Key Inquiry Question: What conditions do green plants require to manufacture food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: What conditions do green plants require to manufacture food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: What conditions do green plants require to manufacture food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 53

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 11	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain processes and requirements involved in plant nutrition.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate the importance of plants and care for the environment.

Key Inquiry Question: What conditions do green plants require to manufacture food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: What conditions do green plants require to manufacture food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: What conditions do green plants require to manufacture food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 54

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 11	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain processes and requirements involved in plant nutrition.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate the importance of plants and care for the environment.

Key Inquiry Question: What conditions do green plants require to manufacture food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: What conditions do green plants require to manufacture food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: What conditions do green plants require to manufacture food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 55

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 1	WEEK: 11	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain processes and requirements involved in plant nutrition.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate the importance of plants and care for the environment.

Key Inquiry Question: What conditions do green plants require to manufacture food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: What conditions do green plants require to manufacture food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: What conditions do green plants require to manufacture food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 56

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 1	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain processes and requirements involved in plant nutrition.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate the importance of plants and care for the environment.

Key Inquiry Question: What conditions do green plants require to manufacture food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: What conditions do green plants require to manufacture food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: What conditions do green plants require to manufacture food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 57

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 1	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain processes and requirements involved in plant nutrition.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate the importance of plants and care for the environment.

Key Inquiry Question: What conditions do green plants require to manufacture food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: What conditions do green plants require to manufacture food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: What conditions do green plants require to manufacture food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 58

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 1	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain processes and requirements involved in plant nutrition.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate the importance of plants and care for the environment.

Key Inquiry Question: What conditions do green plants require to manufacture food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: What conditions do green plants require to manufacture food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: What conditions do green plants require to manufacture food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 59

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 1	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain processes and requirements involved in plant nutrition.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate the importance of plants and care for the environment.

Key Inquiry Question: What conditions do green plants require to manufacture food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: What conditions do green plants require to manufacture food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: What conditions do green plants require to manufacture food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 60

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 1	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain processes and requirements involved in plant nutrition.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate the importance of plants and care for the environment.

Key Inquiry Question: What conditions do green plants require to manufacture food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: What conditions do green plants require to manufacture food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: What conditions do green plants require to manufacture food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 61

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 2	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain processes and requirements involved in plant nutrition.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate the importance of plants and care for the environment.

Key Inquiry Question: What conditions do green plants require to manufacture food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: What conditions do green plants require to manufacture food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: What conditions do green plants require to manufacture food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 62

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 2	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain processes and requirements involved in plant nutrition.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should appreciate the importance of plants and care for the environment.

Key Inquiry Question: What conditions do green plants require to manufacture food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: What conditions do green plants require to manufacture food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: What conditions do green plants require to manufacture food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 63

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 2	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Animals

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate respect for living things and appreciate biological diversity.

Key Inquiry Question: How are animal structures and behaviours adapted to obtaining food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in animals. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How are animal structures and behaviours adapted to obtaining food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How are animal structures and behaviours adapted to obtaining food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 64

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 2	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Animals

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate respect for living things and appreciate biological diversity.

Key Inquiry Question: How are animal structures and behaviours adapted to obtaining food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in animals. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How are animal structures and behaviours adapted to obtaining food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How are animal structures and behaviours adapted to obtaining food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 65

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 2	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Animals

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate respect for living things and appreciate biological diversity.

Key Inquiry Question: How are animal structures and behaviours adapted to obtaining food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in animals. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How are animal structures and behaviours adapted to obtaining food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How are animal structures and behaviours adapted to obtaining food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 66

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 3	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Animals

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate respect for living things and appreciate biological diversity.

Key Inquiry Question: How are animal structures and behaviours adapted to obtaining food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in animals. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How are animal structures and behaviours adapted to obtaining food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How are animal structures and behaviours adapted to obtaining food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 67

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 3	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Animals

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate respect for living things and appreciate biological diversity.

Key Inquiry Question: How are animal structures and behaviours adapted to obtaining food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in animals. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How are animal structures and behaviours adapted to obtaining food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How are animal structures and behaviours adapted to obtaining food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 68

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 3	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Animals

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate respect for living things and appreciate biological diversity.

Key Inquiry Question: How are animal structures and behaviours adapted to obtaining food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in animals. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How are animal structures and behaviours adapted to obtaining food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How are animal structures and behaviours adapted to obtaining food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 69

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 3	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Animals

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate respect for living things and appreciate biological diversity.

Key Inquiry Question: How are animal structures and behaviours adapted to obtaining food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in animals. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How are animal structures and behaviours adapted to obtaining food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How are animal structures and behaviours adapted to obtaining food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 70

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 3	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Animals

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate respect for living things and appreciate biological diversity.

Key Inquiry Question: How are animal structures and behaviours adapted to obtaining food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in animals. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How are animal structures and behaviours adapted to obtaining food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How are animal structures and behaviours adapted to obtaining food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 71

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 4	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Animals

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate respect for living things and appreciate biological diversity.

Key Inquiry Question: How are animal structures and behaviours adapted to obtaining food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in animals. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How are animal structures and behaviours adapted to obtaining food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How are animal structures and behaviours adapted to obtaining food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 72

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 4	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Animals

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate respect for living things and appreciate biological diversity.

Key Inquiry Question: How are animal structures and behaviours adapted to obtaining food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in animals. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How are animal structures and behaviours adapted to obtaining food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How are animal structures and behaviours adapted to obtaining food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 73

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 4	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Animals

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate respect for living things and appreciate biological diversity.

Key Inquiry Question: How are animal structures and behaviours adapted to obtaining food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in animals. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How are animal structures and behaviours adapted to obtaining food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How are animal structures and behaviours adapted to obtaining food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 74

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 4	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Animals

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate respect for living things and appreciate biological diversity.

Key Inquiry Question: How are animal structures and behaviours adapted to obtaining food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in animals. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How are animal structures and behaviours adapted to obtaining food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How are animal structures and behaviours adapted to obtaining food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 75

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 4	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Animals

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate respect for living things and appreciate biological diversity.

Key Inquiry Question: How are animal structures and behaviours adapted to obtaining food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in animals. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How are animal structures and behaviours adapted to obtaining food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How are animal structures and behaviours adapted to obtaining food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 76

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 5	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Animals

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate respect for living things and appreciate biological diversity.

Key Inquiry Question: How are animal structures and behaviours adapted to obtaining food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in animals. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How are animal structures and behaviours adapted to obtaining food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How are animal structures and behaviours adapted to obtaining food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 77

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 5	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Animals

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate respect for living things and appreciate biological diversity.

Key Inquiry Question: How are animal structures and behaviours adapted to obtaining food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in animals. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How are animal structures and behaviours adapted to obtaining food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How are animal structures and behaviours adapted to obtaining food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 78

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 5	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Nutrition in Animals

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate respect for living things and appreciate biological diversity.

Key Inquiry Question: How are animal structures and behaviours adapted to obtaining food?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about nutrition in animals. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How are animal structures and behaviours adapted to obtaining food?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How are animal structures and behaviours adapted to obtaining food? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 79

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 5	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Reproduction in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe processes of sexual and asexual reproduction in plants.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show care for plants and value the role of reproduction in plant survival.

Key Inquiry Question: How do plant structures and processes support reproduction and survival?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about reproduction in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do plant structures and processes support reproduction and survival?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do plant structures and processes support reproduction and survival? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 80

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 5	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Reproduction in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe processes of sexual and asexual reproduction in plants.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show care for plants and value the role of reproduction in plant survival.

Key Inquiry Question: How do plant structures and processes support reproduction and survival?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about reproduction in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do plant structures and processes support reproduction and survival?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do plant structures and processes support reproduction and survival? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 81

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 6	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Reproduction in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe processes of sexual and asexual reproduction in plants.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show care for plants and value the role of reproduction in plant survival.

Key Inquiry Question: How do plant structures and processes support reproduction and survival?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about reproduction in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do plant structures and processes support reproduction and survival?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do plant structures and processes support reproduction and survival? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 82

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 6	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Reproduction in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe processes of sexual and asexual reproduction in plants.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show care for plants and value the role of reproduction in plant survival.

Key Inquiry Question: How do plant structures and processes support reproduction and survival?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about reproduction in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do plant structures and processes support reproduction and survival?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do plant structures and processes support reproduction and survival? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 83

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 6	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Reproduction in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe processes of sexual and asexual reproduction in plants.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show care for plants and value the role of reproduction in plant survival.

Key Inquiry Question: How do plant structures and processes support reproduction and survival?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about reproduction in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do plant structures and processes support reproduction and survival?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do plant structures and processes support reproduction and survival? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 84

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 6	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Reproduction in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe processes of sexual and asexual reproduction in plants.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show care for plants and value the role of reproduction in plant survival.

Key Inquiry Question: How do plant structures and processes support reproduction and survival?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about reproduction in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do plant structures and processes support reproduction and survival?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do plant structures and processes support reproduction and survival? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 85

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 6	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Reproduction in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe processes of sexual and asexual reproduction in plants.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show care for plants and value the role of reproduction in plant survival.

Key Inquiry Question: How do plant structures and processes support reproduction and survival?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about reproduction in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do plant structures and processes support reproduction and survival?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do plant structures and processes support reproduction and survival? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 86

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 7	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Reproduction in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe processes of sexual and asexual reproduction in plants.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show care for plants and value the role of reproduction in plant survival.

Key Inquiry Question: How do plant structures and processes support reproduction and survival?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about reproduction in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do plant structures and processes support reproduction and survival?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do plant structures and processes support reproduction and survival? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 87

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 7	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Reproduction in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe processes of sexual and asexual reproduction in plants.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show care for plants and value the role of reproduction in plant survival.

Key Inquiry Question: How do plant structures and processes support reproduction and survival?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about reproduction in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do plant structures and processes support reproduction and survival?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do plant structures and processes support reproduction and survival? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 88

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 7	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Reproduction in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe processes of sexual and asexual reproduction in plants.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show care for plants and value the role of reproduction in plant survival.

Key Inquiry Question: How do plant structures and processes support reproduction and survival?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about reproduction in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do plant structures and processes support reproduction and survival?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do plant structures and processes support reproduction and survival? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 89

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 7	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Reproduction in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe processes of sexual and asexual reproduction in plants.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show care for plants and value the role of reproduction in plant survival.

Key Inquiry Question: How do plant structures and processes support reproduction and survival?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about reproduction in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do plant structures and processes support reproduction and survival?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do plant structures and processes support reproduction and survival? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 90

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 7	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Reproduction in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe processes of sexual and asexual reproduction in plants.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show care for plants and value the role of reproduction in plant survival.

Key Inquiry Question: How do plant structures and processes support reproduction and survival?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about reproduction in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do plant structures and processes support reproduction and survival?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do plant structures and processes support reproduction and survival? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 91

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 8	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Reproduction in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe processes of sexual and asexual reproduction in plants.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show care for plants and value the role of reproduction in plant survival.

Key Inquiry Question: How do plant structures and processes support reproduction and survival?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about reproduction in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do plant structures and processes support reproduction and survival?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do plant structures and processes support reproduction and survival? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 92

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 8	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Reproduction in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe processes of sexual and asexual reproduction in plants.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show care for plants and value the role of reproduction in plant survival.

Key Inquiry Question: How do plant structures and processes support reproduction and survival?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about reproduction in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do plant structures and processes support reproduction and survival?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do plant structures and processes support reproduction and survival? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 93

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 8	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Reproduction in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe processes of sexual and asexual reproduction in plants.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show care for plants and value the role of reproduction in plant survival.

Key Inquiry Question: How do plant structures and processes support reproduction and survival?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about reproduction in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do plant structures and processes support reproduction and survival?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do plant structures and processes support reproduction and survival? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 94

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 8	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Reproduction in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe processes of sexual and asexual reproduction in plants.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show care for plants and value the role of reproduction in plant survival.

Key Inquiry Question: How do plant structures and processes support reproduction and survival?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about reproduction in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do plant structures and processes support reproduction and survival?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do plant structures and processes support reproduction and survival? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 95

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 8	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Reproduction in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe processes of sexual and asexual reproduction in plants.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show care for plants and value the role of reproduction in plant survival.

Key Inquiry Question: How do plant structures and processes support reproduction and survival?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about reproduction in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do plant structures and processes support reproduction and survival?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do plant structures and processes support reproduction and survival? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 96

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 9	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Reproduction in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe processes of sexual and asexual reproduction in plants.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show care for plants and value the role of reproduction in plant survival.

Key Inquiry Question: How do plant structures and processes support reproduction and survival?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about reproduction in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do plant structures and processes support reproduction and survival?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do plant structures and processes support reproduction and survival? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 97

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 9	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Reproduction in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe processes of sexual and asexual reproduction in plants.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show care for plants and value the role of reproduction in plant survival.

Key Inquiry Question: How do plant structures and processes support reproduction and survival?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about reproduction in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do plant structures and processes support reproduction and survival?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do plant structures and processes support reproduction and survival? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 98

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 9	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Reproduction in Plants

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe processes of sexual and asexual reproduction in plants.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should show care for plants and value the role of reproduction in plant survival.

Key Inquiry Question: How do plant structures and processes support reproduction and survival?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about reproduction in plants. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do plant structures and processes support reproduction and survival?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do plant structures and processes support reproduction and survival? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 99

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 9	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Interdependence of Life

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain relationships among organisms and their environment.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate responsibility for biodiversity and environmental conservation.

Key Inquiry Question: How do organisms depend on one another and on their environment?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about interdependence of life. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do organisms depend on one another and on their environment?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do organisms depend on one another and on their environment? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 100

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 9	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Interdependence of Life

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain relationships among organisms and their environment.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate responsibility for biodiversity and environmental conservation.

Key Inquiry Question: How do organisms depend on one another and on their environment?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about interdependence of life. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do organisms depend on one another and on their environment?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do organisms depend on one another and on their environment? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 101

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 10	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Interdependence of Life

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain relationships among organisms and their environment.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate responsibility for biodiversity and environmental conservation.

Key Inquiry Question: How do organisms depend on one another and on their environment?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about interdependence of life. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do organisms depend on one another and on their environment?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do organisms depend on one another and on their environment? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 102

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 10	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Interdependence of Life

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain relationships among organisms and their environment.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate responsibility for biodiversity and environmental conservation.

Key Inquiry Question: How do organisms depend on one another and on their environment?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about interdependence of life. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do organisms depend on one another and on their environment?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do organisms depend on one another and on their environment? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 103

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 10	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Interdependence of Life

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain relationships among organisms and their environment.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate responsibility for biodiversity and environmental conservation.

Key Inquiry Question: How do organisms depend on one another and on their environment?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about interdependence of life. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do organisms depend on one another and on their environment?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do organisms depend on one another and on their environment? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 104

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 10	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Interdependence of Life

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain relationships among organisms and their environment.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate responsibility for biodiversity and environmental conservation.

Key Inquiry Question: How do organisms depend on one another and on their environment?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about interdependence of life. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do organisms depend on one another and on their environment?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do organisms depend on one another and on their environment? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 105

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 10	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Interdependence of Life

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain relationships among organisms and their environment.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate responsibility for biodiversity and environmental conservation.

Key Inquiry Question: How do organisms depend on one another and on their environment?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about interdependence of life. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do organisms depend on one another and on their environment?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do organisms depend on one another and on their environment? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 106

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 11	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Interdependence of Life

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain relationships among organisms and their environment.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate responsibility for biodiversity and environmental conservation.

Key Inquiry Question: How do organisms depend on one another and on their environment?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about interdependence of life. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do organisms depend on one another and on their environment?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do organisms depend on one another and on their environment? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 107

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 11	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Interdependence of Life

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain relationships among organisms and their environment.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate responsibility for biodiversity and environmental conservation.

Key Inquiry Question: How do organisms depend on one another and on their environment?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about interdependence of life. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do organisms depend on one another and on their environment?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do organisms depend on one another and on their environment? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 108

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 11	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Interdependence of Life

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain relationships among organisms and their environment.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate responsibility for biodiversity and environmental conservation.

Key Inquiry Question: How do organisms depend on one another and on their environment?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about interdependence of life. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do organisms depend on one another and on their environment?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do organisms depend on one another and on their environment? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 109

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 11	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Interdependence of Life

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain relationships among organisms and their environment.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate responsibility for biodiversity and environmental conservation.

Key Inquiry Question: How do organisms depend on one another and on their environment?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about interdependence of life. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do organisms depend on one another and on their environment?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do organisms depend on one another and on their environment? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 110

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 2	WEEK: 11	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Interdependence of Life

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain relationships among organisms and their environment.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate responsibility for biodiversity and environmental conservation.

Key Inquiry Question: How do organisms depend on one another and on their environment?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about interdependence of life. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do organisms depend on one another and on their environment?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do organisms depend on one another and on their environment? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 111

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 1	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Interdependence of Life

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain relationships among organisms and their environment.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate responsibility for biodiversity and environmental conservation.

Key Inquiry Question: How do organisms depend on one another and on their environment?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about interdependence of life. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do organisms depend on one another and on their environment?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do organisms depend on one another and on their environment? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 112

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 1	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Interdependence of Life

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain relationships among organisms and their environment.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate responsibility for biodiversity and environmental conservation.

Key Inquiry Question: How do organisms depend on one another and on their environment?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about interdependence of life. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do organisms depend on one another and on their environment?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do organisms depend on one another and on their environment? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 113

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 1	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Interdependence of Life

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain relationships among organisms and their environment.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate responsibility for biodiversity and environmental conservation.

Key Inquiry Question: How do organisms depend on one another and on their environment?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about interdependence of life. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do organisms depend on one another and on their environment?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do organisms depend on one another and on their environment? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 114

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 1	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Interdependence of Life

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain relationships among organisms and their environment.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate responsibility for biodiversity and environmental conservation.

Key Inquiry Question: How do organisms depend on one another and on their environment?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about interdependence of life. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do organisms depend on one another and on their environment?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do organisms depend on one another and on their environment? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 115

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 1	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Interdependence of Life

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain relationships among organisms and their environment.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate responsibility for biodiversity and environmental conservation.

Key Inquiry Question: How do organisms depend on one another and on their environment?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about interdependence of life. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do organisms depend on one another and on their environment?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do organisms depend on one another and on their environment? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 116

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 2	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Living Things and Environment

Sub-strand: Interdependence of Life

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to explain relationships among organisms and their environment.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate responsibility for biodiversity and environmental conservation.

Key Inquiry Question: How do organisms depend on one another and on their environment?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about interdependence of life. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do organisms depend on one another and on their environment?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do organisms depend on one another and on their environment? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 117

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 2	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Curved Mirrors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should observe safety precautions and show curiosity when investigating light.

Key Inquiry Question: How does the shape of a mirror affect the image formed?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about curved mirrors. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the shape of a mirror affect the image formed?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the shape of a mirror affect the image formed? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 118

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 2	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Curved Mirrors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should observe safety precautions and show curiosity when investigating light.

Key Inquiry Question: How does the shape of a mirror affect the image formed?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about curved mirrors. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the shape of a mirror affect the image formed?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the shape of a mirror affect the image formed? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 119

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 2	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Curved Mirrors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should observe safety precautions and show curiosity when investigating light.

Key Inquiry Question: How does the shape of a mirror affect the image formed?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about curved mirrors. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the shape of a mirror affect the image formed?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the shape of a mirror affect the image formed? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 120

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 2	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Curved Mirrors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should observe safety precautions and show curiosity when investigating light.

Key Inquiry Question: How does the shape of a mirror affect the image formed?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about curved mirrors. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the shape of a mirror affect the image formed?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the shape of a mirror affect the image formed? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 121

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 3	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Curved Mirrors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should observe safety precautions and show curiosity when investigating light.

Key Inquiry Question: How does the shape of a mirror affect the image formed?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about curved mirrors. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the shape of a mirror affect the image formed?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the shape of a mirror affect the image formed? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 122

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 3	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Curved Mirrors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should observe safety precautions and show curiosity when investigating light.

Key Inquiry Question: How does the shape of a mirror affect the image formed?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about curved mirrors. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the shape of a mirror affect the image formed?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the shape of a mirror affect the image formed? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 123

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 3	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Curved Mirrors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should observe safety precautions and show curiosity when investigating light.

Key Inquiry Question: How does the shape of a mirror affect the image formed?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about curved mirrors. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the shape of a mirror affect the image formed?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the shape of a mirror affect the image formed? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 124

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 3	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Curved Mirrors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should observe safety precautions and show curiosity when investigating light.

Key Inquiry Question: How does the shape of a mirror affect the image formed?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about curved mirrors. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the shape of a mirror affect the image formed?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the shape of a mirror affect the image formed? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 125

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 3	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Curved Mirrors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should observe safety precautions and show curiosity when investigating light.

Key Inquiry Question: How does the shape of a mirror affect the image formed?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about curved mirrors. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the shape of a mirror affect the image formed?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the shape of a mirror affect the image formed? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 126

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 4	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Curved Mirrors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should observe safety precautions and show curiosity when investigating light.

Key Inquiry Question: How does the shape of a mirror affect the image formed?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about curved mirrors. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the shape of a mirror affect the image formed?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the shape of a mirror affect the image formed? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 127

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 4	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Curved Mirrors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should observe safety precautions and show curiosity when investigating light.

Key Inquiry Question: How does the shape of a mirror affect the image formed?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about curved mirrors. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the shape of a mirror affect the image formed?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the shape of a mirror affect the image formed? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 128

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 4	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Curved Mirrors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should observe safety precautions and show curiosity when investigating light.

Key Inquiry Question: How does the shape of a mirror affect the image formed?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about curved mirrors. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the shape of a mirror affect the image formed?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the shape of a mirror affect the image formed? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 129

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 4	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Curved Mirrors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should observe safety precautions and show curiosity when investigating light.

Key Inquiry Question: How does the shape of a mirror affect the image formed?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about curved mirrors. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the shape of a mirror affect the image formed?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the shape of a mirror affect the image formed? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 130

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 4	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Curved Mirrors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should observe safety precautions and show curiosity when investigating light.

Key Inquiry Question: How does the shape of a mirror affect the image formed?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about curved mirrors. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the shape of a mirror affect the image formed?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the shape of a mirror affect the image formed? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 131

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 5	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Curved Mirrors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should observe safety precautions and show curiosity when investigating light.

Key Inquiry Question: How does the shape of a mirror affect the image formed?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about curved mirrors. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the shape of a mirror affect the image formed?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the shape of a mirror affect the image formed? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 132

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 5	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Curved Mirrors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should observe safety precautions and show curiosity when investigating light.

Key Inquiry Question: How does the shape of a mirror affect the image formed?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about curved mirrors. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the shape of a mirror affect the image formed?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the shape of a mirror affect the image formed? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 133

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 5	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Curved Mirrors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should observe safety precautions and show curiosity when investigating light.

Key Inquiry Question: How does the shape of a mirror affect the image formed?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about curved mirrors. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the shape of a mirror affect the image formed?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the shape of a mirror affect the image formed? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 134

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 5	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Curved Mirrors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should observe safety precautions and show curiosity when investigating light.

Key Inquiry Question: How does the shape of a mirror affect the image formed?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about curved mirrors. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How does the shape of a mirror affect the image formed?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How does the shape of a mirror affect the image formed? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 135

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 5	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Waves

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe wave characteristics and investigate how waves transfer energy.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate curiosity, cooperation and safe participation during investigations.

Key Inquiry Question: How do waves transfer energy from one place to another?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about waves. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do waves transfer energy from one place to another?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do waves transfer energy from one place to another? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 136

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 6	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Waves

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe wave characteristics and investigate how waves transfer energy.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate curiosity, cooperation and safe participation during investigations.

Key Inquiry Question: How do waves transfer energy from one place to another?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about waves. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do waves transfer energy from one place to another?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do waves transfer energy from one place to another? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 137

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 6	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Waves

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe wave characteristics and investigate how waves transfer energy.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate curiosity, cooperation and safe participation during investigations.

Key Inquiry Question: How do waves transfer energy from one place to another?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about waves. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do waves transfer energy from one place to another?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do waves transfer energy from one place to another? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 138

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 6	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Waves

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe wave characteristics and investigate how waves transfer energy.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate curiosity, cooperation and safe participation during investigations.

Key Inquiry Question: How do waves transfer energy from one place to another?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about waves. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do waves transfer energy from one place to another?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do waves transfer energy from one place to another? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 139

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 6	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Waves

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe wave characteristics and investigate how waves transfer energy.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate curiosity, cooperation and safe participation during investigations.

Key Inquiry Question: How do waves transfer energy from one place to another?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about waves. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do waves transfer energy from one place to another?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do waves transfer energy from one place to another? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 140

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 6	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Waves

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe wave characteristics and investigate how waves transfer energy.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate curiosity, cooperation and safe participation during investigations.

Key Inquiry Question: How do waves transfer energy from one place to another?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about waves. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do waves transfer energy from one place to another?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do waves transfer energy from one place to another? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 141

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 7	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Waves

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe wave characteristics and investigate how waves transfer energy.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate curiosity, cooperation and safe participation during investigations.

Key Inquiry Question: How do waves transfer energy from one place to another?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about waves. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do waves transfer energy from one place to another?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do waves transfer energy from one place to another? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 142

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 7	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Waves

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe wave characteristics and investigate how waves transfer energy.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate curiosity, cooperation and safe participation during investigations.

Key Inquiry Question: How do waves transfer energy from one place to another?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about waves. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do waves transfer energy from one place to another?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do waves transfer energy from one place to another? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 143

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 7	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Waves

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe wave characteristics and investigate how waves transfer energy.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate curiosity, cooperation and safe participation during investigations.

Key Inquiry Question: How do waves transfer energy from one place to another?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about waves. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do waves transfer energy from one place to another?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do waves transfer energy from one place to another? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 144

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 7	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Waves

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe wave characteristics and investigate how waves transfer energy.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate curiosity, cooperation and safe participation during investigations.

Key Inquiry Question: How do waves transfer energy from one place to another?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about waves. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do waves transfer energy from one place to another?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do waves transfer energy from one place to another? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 145

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 7	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Waves

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe wave characteristics and investigate how waves transfer energy.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate curiosity, cooperation and safe participation during investigations.

Key Inquiry Question: How do waves transfer energy from one place to another?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about waves. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do waves transfer energy from one place to another?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do waves transfer energy from one place to another? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 146

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 8	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Waves

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe wave characteristics and investigate how waves transfer energy.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate curiosity, cooperation and safe participation during investigations.

Key Inquiry Question: How do waves transfer energy from one place to another?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about waves. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do waves transfer energy from one place to another?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do waves transfer energy from one place to another? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 147

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 8	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Waves

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe wave characteristics and investigate how waves transfer energy.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate curiosity, cooperation and safe participation during investigations.

Key Inquiry Question: How do waves transfer energy from one place to another?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about waves. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do waves transfer energy from one place to another?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do waves transfer energy from one place to another? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 148

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 8	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Waves

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe wave characteristics and investigate how waves transfer energy.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate curiosity, cooperation and safe participation during investigations.

Key Inquiry Question: How do waves transfer energy from one place to another?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about waves. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do waves transfer energy from one place to another?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do waves transfer energy from one place to another? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 149

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 8	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Waves

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe wave characteristics and investigate how waves transfer energy.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate curiosity, cooperation and safe participation during investigations.

Key Inquiry Question: How do waves transfer energy from one place to another?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about waves. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do waves transfer energy from one place to another?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do waves transfer energy from one place to another? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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GRADE 9 INTEGRATED SCIENCE – LESSON PLAN 150

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Integrated Science	DATE: _____
TERM: 3	WEEK: 8	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Force and Energy

Sub-strand: Waves

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe wave characteristics and investigate how waves transfer energy.
- **Skills:** The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence.
- **Attitude:** The learner should demonstrate curiosity, cooperation and safe participation during investigations.

Key Inquiry Question: How do waves transfer energy from one place to another?

Learning Resources: KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups. Practical investigations are conducted under appropriate supervision and safety procedures. Learners observe, predict, investigate, measure where relevant, record evidence, discuss findings and communicate conclusions.

Lesson Development

5 minutes – Introduction

Link the lesson to learners' everyday experiences and prior scientific knowledge about waves. Present a short phenomenon, specimen, model, question or demonstration. Learners make observations, share predictions and identify what they want to find out. Introduce the key inquiry question: How do waves transfer energy from one place to another?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.

Step 1: Learners observe the relevant phenomenon, specimen, diagram or model and share prior ideas and predictions.

Step 2: In pairs or groups, learners carry out the investigation, demonstration, model-building or guided task described in the scheme, observing all relevant safety precautions.

Step 3: Learners record observations, measurements, diagrams or results in their notebooks and compare evidence with their initial predictions.

Step 4: Learners discuss findings, answer the key inquiry question and present evidence-based explanations to peers; they correct misconceptions using feedback.

Extended Activities

Learners who complete the core task investigate an additional example or real-life application, prepare a labelled diagram/model, or formulate a question for classmates. Learners needing support revisit key vocabulary and use guided diagrams, concrete materials, peer support or a simplified observation table. Any practical extension must remain age-appropriate and follow school laboratory and fieldwork safety procedures.

5 minutes – Conclusion

Learners summarise the key scientific idea and respond to the inquiry question: How do waves transfer energy from one place to another? They complete a short exit question, label a diagram, state one observation or explain one conclusion supported by evidence.

Assessment

Suggested assessment evidence: Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of practical and scientific skills, misconceptions or safety issues observed, learners requiring support, and the next instructional step.

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Implementation Notes

This collection contains 150 Grade 9 Integrated Science lesson plans organised according to the three-term full-year scheme used to prepare this resource. The strand, sub-strand, lesson outcomes, key inquiry questions, learning resources and assessment methods are drawn from that scheme.

Each lesson is structured for a 40-minute period: 5 minutes introduction, 30 minutes lesson development and 5 minutes conclusion. Adjust timing to the school's timetable, the nature of practical work, learner progress and available facilities.

Practical activities must be conducted with suitable supervision, appropriate protective equipment and the school's laboratory safety procedures. Do not undertake an activity where the required safe conditions or equipment are unavailable.

This is an original teacher-prepared resource aligned to the Grade 9 curriculum design; it is not an official KICD publication. Verify the sequence and school calendar before classroom implementation.